AI / COMPUTER VISION PROJECT SUBMISSION

YOLO Real-Time Object Detection using Webcam

Deep Learning Based Live Object Detection System

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Table of Contents

Project Overview

This project implements a real-time object detection system using the YOLO (You Only Look Once) deep learning algorithm with a live webcam feed. The system detects multiple objects simultaneously, draws bounding boxes around each detected object, and displays the corresponding class labels along with confidence scores in real time.

The project demonstrates the efficiency of YOLO for fast object detection by processing each frame in a single forward pass, making it suitable for real-time applications such as surveillance systems, robotics, automation, and intelligent monitoring.

Key Features

- Real-time object detection using webcam
- Detects multiple objects simultaneously
- Displays bounding boxes, labels, and confidence scores
- Uses YOLOv3 and YOLOv3-Tiny models
- Adjustable confidence threshold
- Efficient detection using the OpenCV DNN module
- Uses the COCO dataset for object classes

Technologies & Tools Used

Programming Language	Python 3.11
Libraries	OpenCV (DNN Module), NumPy
Deep Learning Model	YOLOv3, YOLOv3-Tiny
Dataset	COCO Dataset
Development Tools	Git, GitHub, Git LFS

How the Project Works

1. The webcam continuously captures live video frames.
2. Each frame is passed into the YOLO neural network.
3. YOLO divides the frame into grids and predicts object locations and classes.
4. Bounding boxes with confidence scores are generated.
5. Non-Maximum Suppression removes duplicate, overlapping detections.

6 Final detections are displayed in real time with labels and confidence scores.

My Role & Contributions

I independently designed and implemented the complete project. My responsibilities included:

- Developing the real-time object detection pipeline
- Integrating YOLO with OpenCV
- Loading and managing YOLO configuration and weight files
- Processing webcam frames in real time
- Implementing object detection and visualization
- Displaying bounding boxes, labels, and confidence scores
- Organizing the project structure
- Managing the project repository using Git and GitHub
- Writing project documentation (README)

Project Structure

```
YOLO-Real-Time-Object-Detection-Webcam/
|
|-- coco.names
|-- yolov3.cfg
|-- yolov3-tiny.cfg
|-- real_time_yolo_webcam.py
|-- real_time_yolo_detector1.py
|-- real_time_yolo_detector2.py
`-- README.md
```

GitHub Repository

The complete source code for this project is available at:

https://github.com/Shikha18Shukla/YOLO-Real-Time-Object-Detection-Webcam-

Project Implementation

The following screenshots capture the system in operation, showing live webcam frames with YOLO-generated bounding boxes, class labels, and confidence scores.

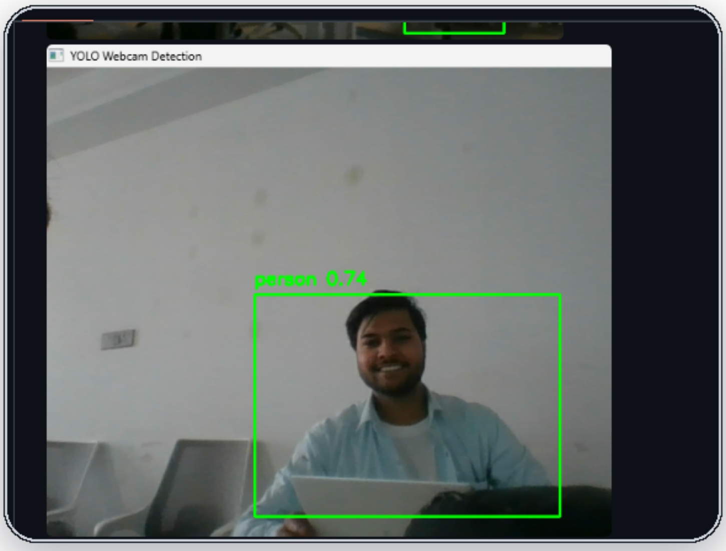

Figure 1: Real-Time Object Detection Output

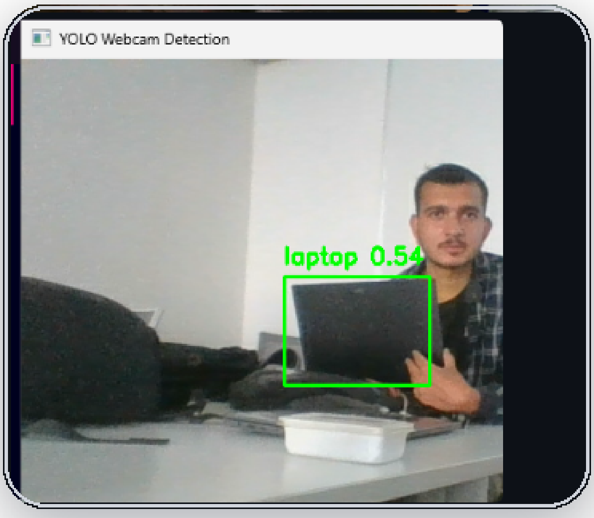

Figure 2: Webcam Detection Interface

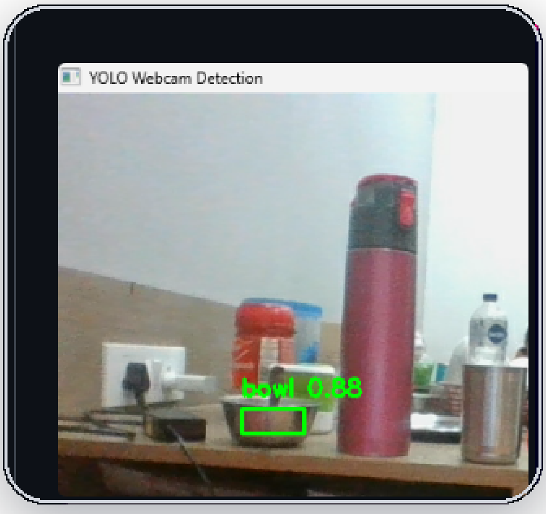

Figure 3: Multi-Class Object Recognition

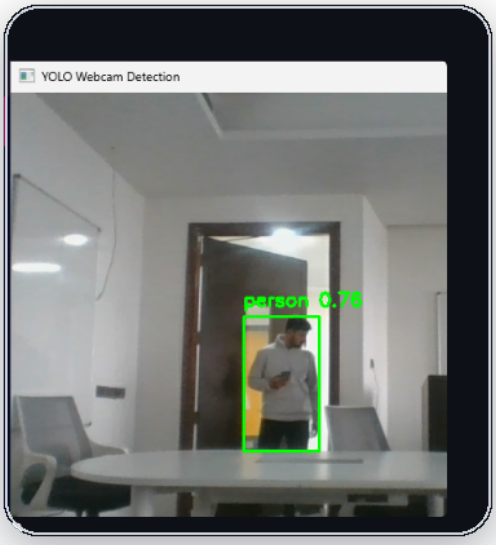

Figure 4: Real-Time Person Detection

Project Applications

Smart Surveillance	**Traffic Monitoring**
Autonomous Vehicles	**Robotics**
Industrial Automation	**Smart Security Systems**
Inventory Monitoring	**Intelligent Video Analytics**

Learning Outcomes

- Fundamentals of Computer Vision
- Deep Learning-based Object Detection
- Working with YOLO architecture
- Image preprocessing
- OpenCV DNN module
- Real-time video processing
- Confidence threshold tuning
- Non-Maximum Suppression
- Git and GitHub project management

Conclusion

This project demonstrates the practical implementation of real-time object detection using the YOLO deep learning algorithm. It strengthened my understanding of computer vision, deep learning inference, and real-time image processing while providing hands-on experience with industry-relevant AI technologies.